Satellite Image Analysis (EuroSAT)

Land Use & Land Cover Classification Technical Report

Dataset: EuroSAT (RGB 64x64) | Architectures: ViT, Swin Transformer, ViT-Swin Hybrid

1. Project Overview

1.1 Purpose of the Graduation Project

The primary objective of this graduation project is to design, implement, and evaluate an automated Computer Vision framework for Satellite Image Analysis. By focusing on Land Use and Land Cover (LULC) classification using Sentinel-2 RGB satellite imagery, the system aims to automatically categorize satellite patches into 10 distinct environmental and infrastructure classes using state-of-the-art Vision Transformers.

1.2 Expected AI Development Lifecycle

The project adheres strictly to a systematic AI engineering lifecycle:

- **Problem Definition & Scoping:** Formulating the business problem, expected impact, and quantitative evaluation criteria.
- **Data Acquisition & Preparation:** Acquiring and structuring the public EuroSAT dataset (27,000 images).
- **Exploratory Data Analysis (EDA):** Analyzing class distribution, spatial attributes, and potential spectral challenges.
- **Preprocessing & Augmentation:** Applying image resizing, RGB normalization, and spatial augmentations.
- **Data Partitioning:** Constructing independent Train (70%), Validation (15%), and Test (15%) subsets.
- **Model Engineering:** Implementing Vision Transformer (ViT), Swin Transformer, and a feature-fusion Hybrid architecture.
- **Comprehensive Evaluation:** Evaluating generalization on unseen Test data using multi-class metrics (Accuracy, Precision, Recall, F1-Score, ROC-AUC) and Confusion Matrices.

2. General Project Deliverables

2.1 Problem Understanding

- **Real-World Problem:** Manual ground surveys and conventional remote sensing techniques for land monitoring are extremely labor-intensive, costly, and inefficient when scaling across large geographic areas.
- **Motivation & Expected Impact:** Automated LULC classification enables government authorities, urban planners, and environmental protection agencies to track urban expansion, agricultural management, deforestation, and climate change effects dynamically.

- **Objectives & Success Criteria:** The objective is to develop and compare three deep learning approaches for multi-class Land Use and Land Cover (LULC) classification: Vision Transformer (ViT), Swin Transformer, and a ViT-Swin Hybrid model. Model performance is evaluated on an unseen test set using Accuracy, Precision, Recall, F1-Score, ROC-AUC, and Confusion Matrices. A test accuracy above 90% is defined as the target performance threshold.

2.2 Dataset Preparation

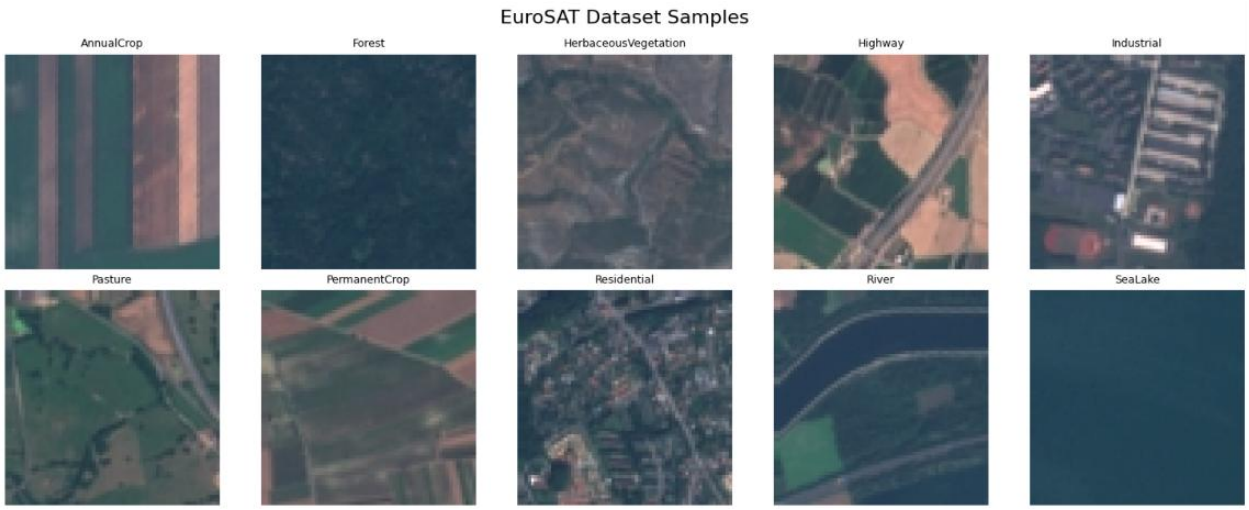
- **Dataset URL:** EuroSAT: A Novel Dataset and Deep Learning Benchmark for Land Use and Land Cover Classification | Zenodo
- **Selected Dataset:** EuroSAT (RGB version), containing 27,000 image patches collected by Sentinel-2 satellites across 13 European countries.
- **Preprocessing:** RGB images were resized to 64 × 64 pixels and normalized using ImageNet channel-wise statistics (mean = [0.485, 0.456, 0.406], std = [0.229, 0.224, 0.225]). Training images were augmented using random horizontal and vertical flips. Validation and test images were resized and normalized without random augmentation to ensure consistent evaluation.
- **Data Augmentation:** To enhance model generalization and mitigate overfitting, spatial transformations including Random Horizontal Flip and Random Vertical Flip were applied to training samples.
- **Data Partitioning:** The dataset was randomly divided using a fixed random seed (42) into training (70%; 18,900 images), validation (15%; 4,050 images), and test (15%; 4,050 images). The split was not stratified, so class proportions may vary slightly across subsets.

2.3 Exploratory Data Analysis (EDA)

- **Dataset Statistics:** 27,000 RGB images with native resolution of 64x64 pixels across 10 balanced categories.
- **Class Distribution:** The EuroSAT RGB dataset contains 27,000 images across 10 land-use and land-cover classes. The number of images per class ranges from 2,000 to 3,000. Although the dataset is relatively balanced, the class frequencies are not identical. This variation should be considered when interpreting per-class evaluation metrics.

Class	Distribution
Annual Crop	3000
Forest	3000
Herbaceous Vegetation	3000
Highway	2500
Industrial	2500

Pasture	2000
Permanent Crop	2500
Residential	3000
River	2500
Sea Lake	3000



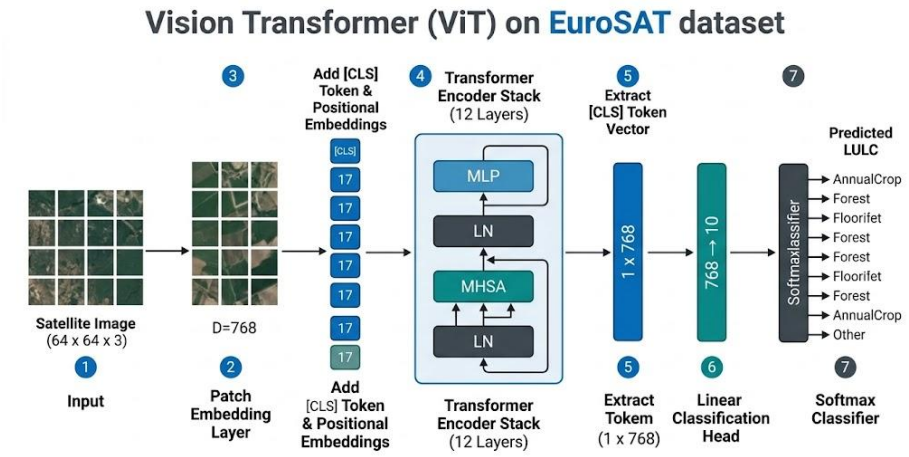
- **Data Challenges & Limitations:** Low spatial resolution (64x64) poses visual fine-grained challenges, particularly in distinguishing spectrally similar classes such as Annual Crop vs. Permanent Crop, or Highway vs. River.

2.4 Model Development

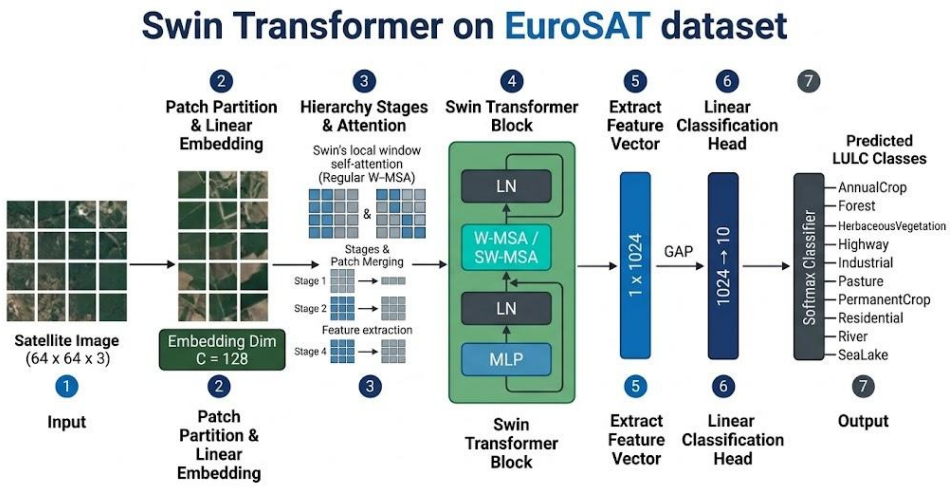
- **Hyperparameter Tuning Experiments**: Three experimental configurations were evaluated to investigate the effect of learning rate and weight decay on model performance. The experiments used validation performance to guide model selection. The final configuration and the corresponding validation results are reported in the experiment comparison table.
  The ViT training pipeline used the AdamW optimizer, a ReduceLROnPlateau learning-rate scheduler (factor = 0.5, patience = 2), and Automatic Mixed Precision (AMP) where supported.

Experiment	Learning Rate	Weight Decay	Best Validation Accuracy
Experiment 1	1e-4	1e-4	97.14%
Experiment 2	1e-3	1e-4	60.40%
Experiment 3	1e-4	1e-3	96.44%

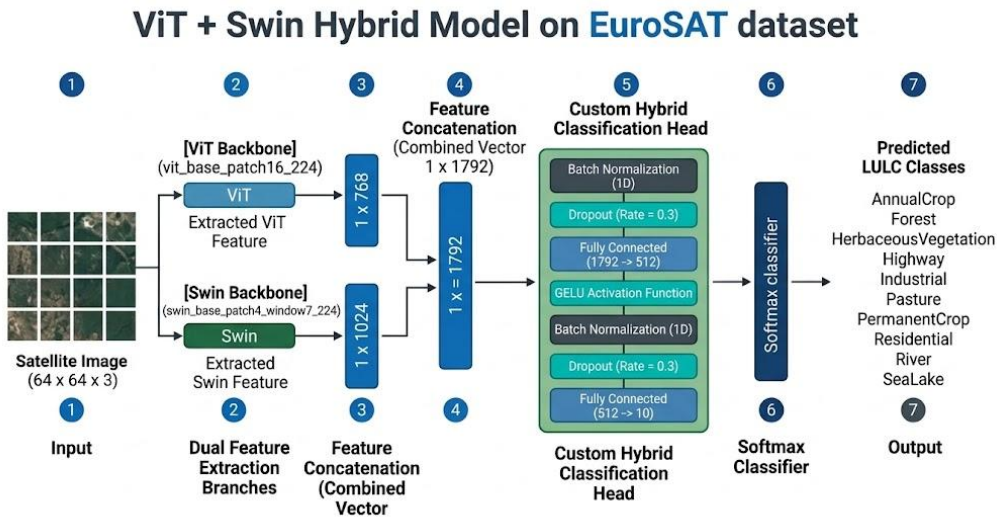
- **Vision Transformer (ViT):** Uses patch embeddings and global self-attention to learn image representations for multi-class classification. The model was initialized with pretrained weights and fine-tuned on the EuroSAT RGB dataset.



- **Swin Transformer:** Uses hierarchical feature representations and shifted-window self-attention to capture local and multi-scale visual patterns.



- **ViT + Swin Hybrid Model:** The hybrid architecture combines a Vision Transformer (ViT) backbone and a Swin Transformer backbone initialized from previously fine-tuned checkpoints. The classification heads of both backbones are replaced with identity layers to obtain their feature representations. These features are concatenated and passed to a classification head consisting of Batch Normalization, Dropout (0.3), a fully connected layer with 512 units, GELU activation, another Batch Normalization and Dropout layer, and a final 10-class output layer. During hybrid training, the backbone parameters and the classification head are optimized jointly using AdamW. The model is trained for 15 epochs, with the best checkpoint selected according to validation accuracy.



2.5 Model Evaluation & Comparative Analysis

All three models were trained for 15 epochs on the training split, checkpointed based on Validation performance, and finally evaluated on the completely unseen Test Set (4,050 samples).

Evaluation Metric	Vision Transformer (ViT)	Swin Transformer	ViT + Swin Hybrid
Test Accuracy	97.14%	98.27%	98.42%
Precision (Macro)	0.9712	0.9823	0.9843
Recall (Macro)	0.9708	0.9822	0.9835
F1-Score (Macro)	0.9709	0.9822	0.9838
Precision (Weighted)	0.9717	0.9827	0.9843

Recall (Weighted)	0.9714	0.9827	0.9842
F1-Score (Weighted)	0.9714	0.9827	0.9842
ROC-AUC (OVR)	0.9992	0.9997	0.9993

2.6 Discussion & Error Analysis

The experimental evaluation demonstrates that combining ViT and Swin Transformer into a Hybrid model achieves superior predictive robustness compared to individual backbones. The hybrid approach effectively bridges global contextual understanding (from ViT) with localized window-based spatial representations (from Swin), resulting in lower confusion between spectrally similar classes such as Highway vs. River and PermanentCrop vs. AnnualCrop.

3. Future AI Application Deployment

To demonstrate practical utility, a Streamlit-based interactive web application interface is designed to ingest input satellite image patches, execute live inference using the fine-tuned Hybrid model checkpoint (best_eurosat_hybrid.pth), and display predicted LULC class labels alongside predicted class probability distribution graphs.

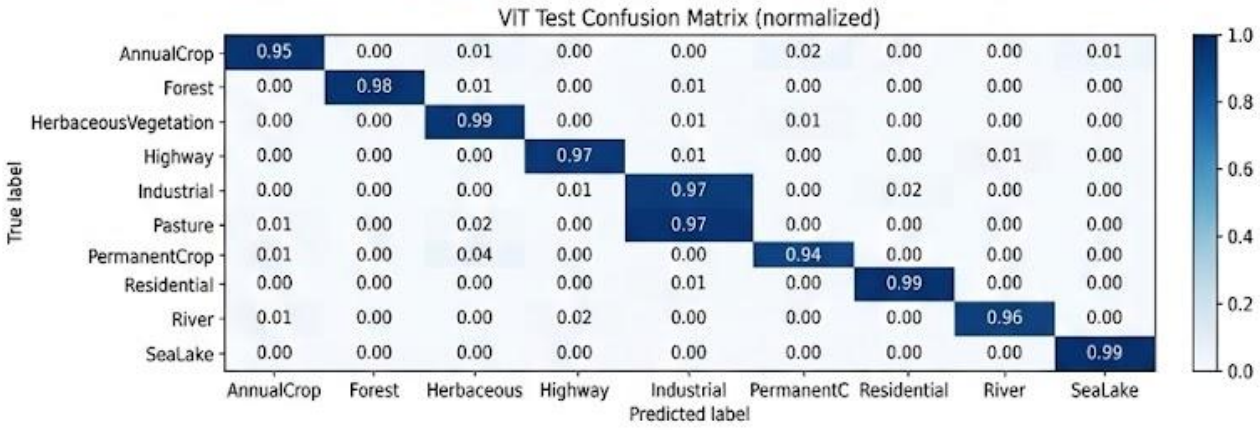
4. Technical Deliverables Summary

- **Source Code:** Self-contained PyTorch Colab notebooks for ViT, Swin, and Hybrid training pipelines.
- **Confusion Matrices:** High-resolution normalized Confusion Matrix plots saved for each model on the Test set.

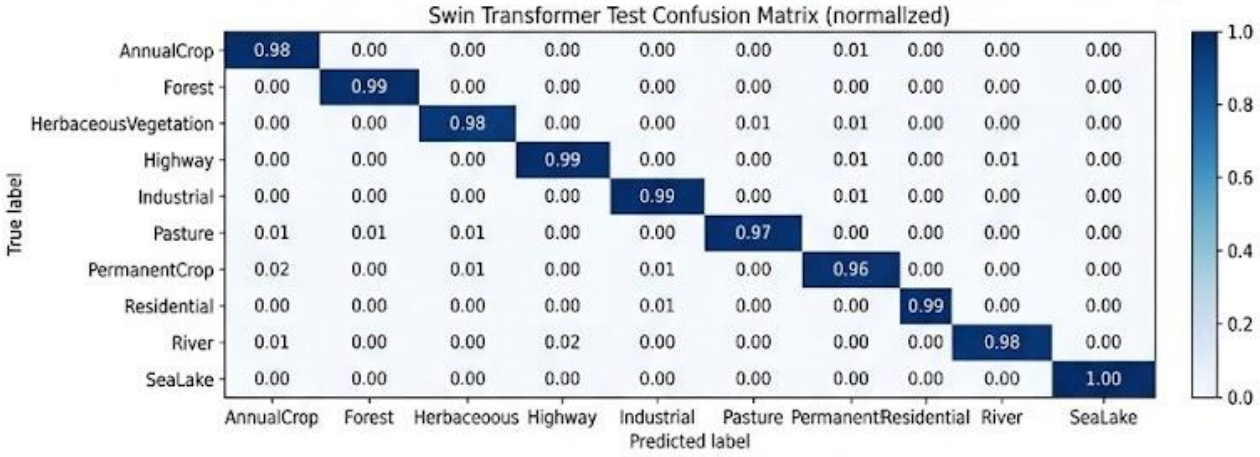
EuroSAT Land Use & Land Cover Classification Model Comparison

Visualizing model performance on the unseen Test Set (4,050 samples).

Vision Transformer (ViT) - 1. Baseline Model (Test Confusion Matrix)



Swin Transformer - 2. Advanced Architecture (Test Confusion Matrix)



ViT + Swin Hybrid - 3. Ensembled Model (Test Confusion Matrix)

